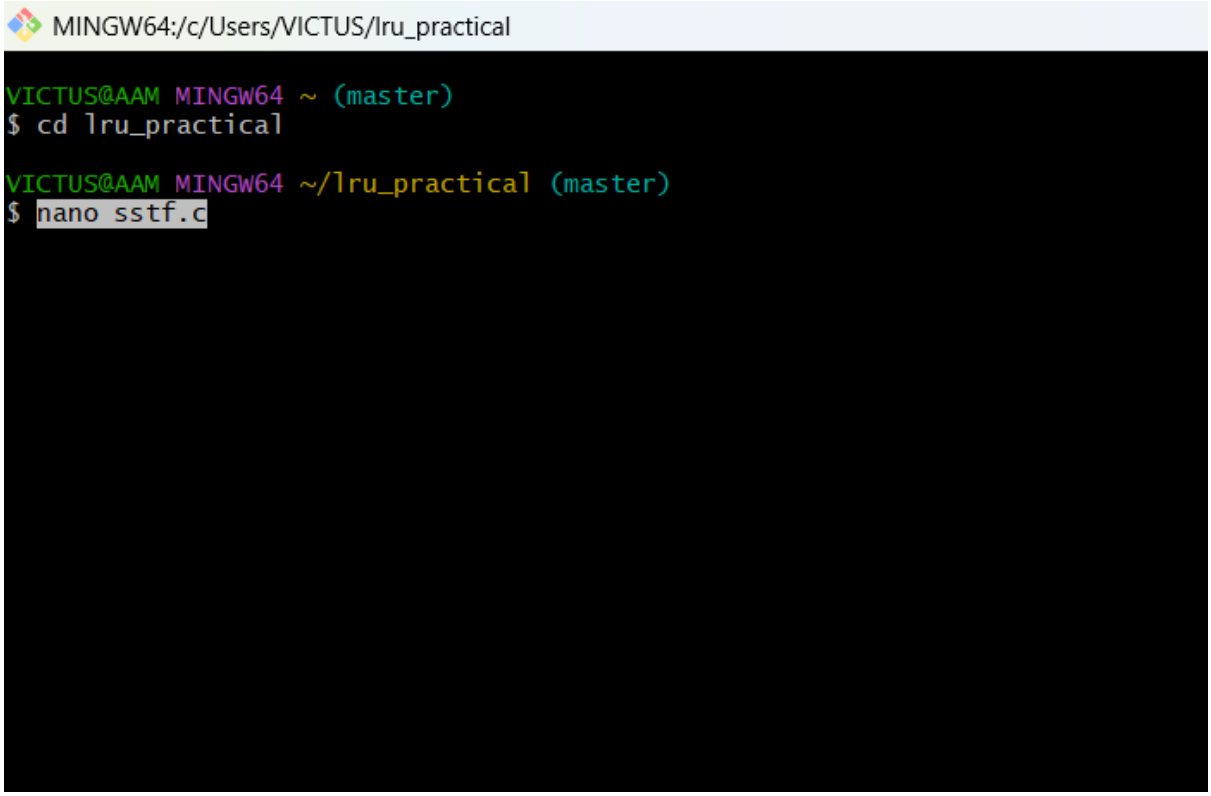
Practical No. 10

Aim

To write and execute a program to calculate the ***total head movement using SSTF (Shortest Seek Time First) disk scheduling algorithm***

Step 1: *Navigate to Directory and Create File*

The working directory was opened and a new C file was created using nano editor:



Step 2: Write Program

The SSTF disk scheduling program was written in C language with:

- *Disk requests = 98, 183, 41, 122, 14, 124, 65, 67*
- *Initial head position = 53*

The program selects the request with the minimum distance from the current head position and calculates the total head movement.

MINGW64:/c/Users/VICTUS/lru_practical

GNU nano 8.7

```c
#include <stdio.h>
#include <stdlib.h>

int main() {
    int requests[] = {98, 183, 41, 122, 14, 124, 65, 67};
    int n = 8;
    int head = 53;

    int visited[8] = {0};
    int total_movement = 0;

    for(int i = 0; i < n; i++) {
        int min_dist = 1000;
        int index = -1;

        for(int j = 0; j < n; j++) {
            if(!visited[j]) {
                int dist = abs(head - requests[j]);
                if(dist < min_dist) {
                    min_dist = dist;
                    index = j;
                }
            }
        }

        visited[index] = 1;
        total_movement += min_dist;
        head = requests[index];
    }

    printf("Total Head Movement = %d\n", total_movement);

    return 0;
}
```

Step 3: Compile and Execute Program

The program was compiled and executed using:

MINGW64:/c/Users/VICTUS/lru_practical

```
VICTUS@AAM MINGW64 ~ (master)
$ cd lru_practical

VICTUS@AAM MINGW64 ~/lru_practical (master)
$ nano sstf.c

VICTUS@AAM MINGW64 ~/lru_practical (master)
$ gcc sstf.c -o sstf
./sstf
```

Step 5: Output Observed

The output obtained is:

- *Total Head Movement = 323*

MINGW64:/c/Users/VICTUS/lru_practical

```
VICTUS@AAM MINGW64 ~ (master)
$ cd lru_practical

VICTUS@AAM MINGW64 ~/lru_practical (master)
$ nano sstf.c

VICTUS@AAM MINGW64 ~/lru_practical (master)
$ gcc sstf.c -o sstf
./sstf
Total Head Movement = 323

VICTUS@AAM MINGW64 ~/lru_practical (master)
$
```

Result

The SSTF disk scheduling program was successfully executed and the total head movement was calculated correctly.